appendix C
Clients

One of the strengths of Elasticsearch is the rich interfacing mechanisms provided out of the box for many clients to perform search and aggregations. Whether we use a Java, Python, C#, JavaScript, or other main programming language client that supports the RESTful API over HTTP protocol, there's always first-class support.

Exposing APIs over a RESTful interface is a modern architectural decision that helps to create language-agnostic products. Elastic embraced this path to create a product that can be integrated with multi-language landscapes.

Currently, Elasticsearch can be integrated with these clients: Java, Python, .NET, Ruby, Go, JavaScript, Perl, and PHP, as well as community-contributed clients such as C++, Kotlin, Scala, Swift, Rust, Haskell, Erlang, and a few others. Elasticsearch is more than happy to consider a request for a bespoke client should the need arise. Put in a pull request to the team, and get it in motion.

Discussing all these clients is impractical (and I do not have enough experience with languages other than Java, Kotlin, Scala, Python, Go, JavaScript, and a couple of others). So, this appendix provides high-level query mechanics for Java, JavaScript, and Python.

C.1 Java client

Elasticsearch is written in Java, and as you may expect, Elasticsearch offers native support for invoking Elasticsearch APIs using the Java Client library. It is built using fluent API builder patterns and supports both synchronous and asynchronous (blocking and nonblocking) API invocations. It requires Java 8 at minimum, so make sure your applications are at least on Java 8.

In the next couple of sections, we look at Elasticsearch integration from a Maven/Gradle Java-based application as well as a Spring-based Java application.